Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/23/2006
PI Name:	Ellis, Stephen Ph.D.		
Project Title:	Kinesthetic Compensation for Sensorimotor Rearrangement		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Space human factors engineering		
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Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
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Space Biology Special Category:	None		
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Zip Code:	94035	**Congressional District:**	14
Comments:			
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No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
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Key Personnel Changes/Previous PI:	Ellis, Stephen R.		
COI Name (Institution):			
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Performance Goal No.:			
Performance Goal Text:			
	Laboratory studies will be conducted to evaluate a newly proposed technique to allow users to control a computer screen cursor in the presence of large rotational sensorimotor rearrangements. We call this technique "kinesthetic cueing." Such rearrangements commonly occur when users teleoperate manipulators while remote cameras provide views of the worksite which are rotated with respect to the direction of command action. If the direction and position of the camera were sensed, consistent display and control actions can be achieved computationally. This remapping is, however, not always possible and users of teleoperations systems often must train to work in rotated coordinate systems. Development of portable, body-worn information appliances such as head-mounted see-through and partially occlusive displays has introduced a new environment in which rotated coordinate system can interfere with system usability. These portable systems often have Graphical User Interfaces for which the user inputs spatially continuous information with a cursor control device. Since these appliances are portable, the cursor control device must be hand-held or body mounted. Our main question is whether kinesthetic information from holding or body mounting the cursor control		

Task Description:	device assists users who operate under conditions of rotational sensorimotor rearrangement. Kinesthetic cueing will be provided to our experimental users by having them contact the cursor control device with the hand not controlling the cursor. This hand, usually the nondominant left hand, will be rotated to copy the rotation, i.e., misalignment angle, with which the operator must deal. Preserving its rotated position, it is then placed in con-tact with the cursor control device. Informal observations suggest that the proposed technique works. Experiment 1 is intended to formally test the idea and determine whether the cueing hand needs to be the nondominant hand. If this technique can be successfully developed, it will solve two problems associated with training users to deal with sensorimotor rearrangement. First, it could reduce or eliminate the need for adaptation training. Secondly, it could remove concern about the perceptual aftereffects of adaptation. To examine this second possibility, Experiment 2 is specifically intended to measure the size of possible perceptual aftereffects experienced by users whose training to operate under conditions of sensorimotor rearrangement has been assisted by kinesthetic cueing. Finally, Experiment 3 will investigate the benefits of kinesthetic cueing with a variety of cursor input devices besides the graphics tablet used in Experiment 1. These will include trackpads, isometric joysticks, mice and tablet devices. Early results of these three experiments will be provided to the developers of the Wireless Augmented Reality Prototype (WARP) Phase III, a head-mounted, wireless display system being designed at NASA Jet Propulsion Laboratory (JPL). The design of WARP's cursor controller will be advanced by appropriate kines-thetic cueing. Benefits are also expected in the design of human interfaces for teleoperations.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The improved human interfaces for telerobotics being developed for Shuttle and Space Station applications are directly applicable to telesurgery, laparoscopic and endoscopic surgery as well as all other forms of remote control.
Task Progress:	No progress report this reporting period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)